

CSE 4417 IoT Theory

Lecture 10: IoT Security

Atanu Shuvam Roy



Lecturer,
Dept. of CSE, ULAB, Dhaka



EMAIL:

atanu.shuvam@ulab.edu.bd



Class Code
bjedchaj



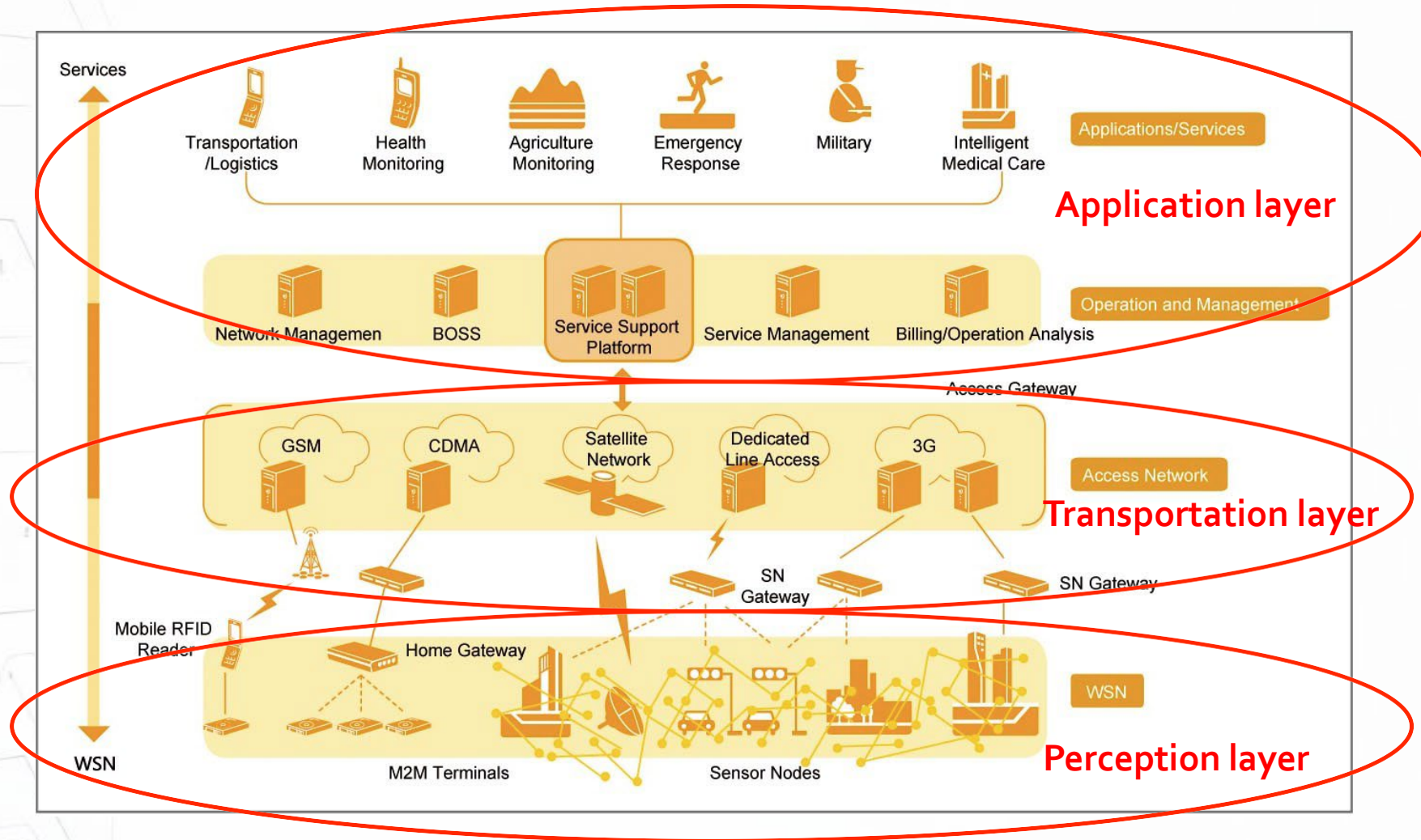
Spring 2026

Security and Privacy in the IoT

- What is security?
 - Who is accessing the data?
 - **Protect the data**
- What is privacy?
 - Whose data is being accessed?
 - **Protect the user**
- Issues with the IoT systems:
 - Hardware heterogeneity
 - Limited resources
 - Big data generation
 - Need for complex data protection and access measures



Hierarchical IoT System

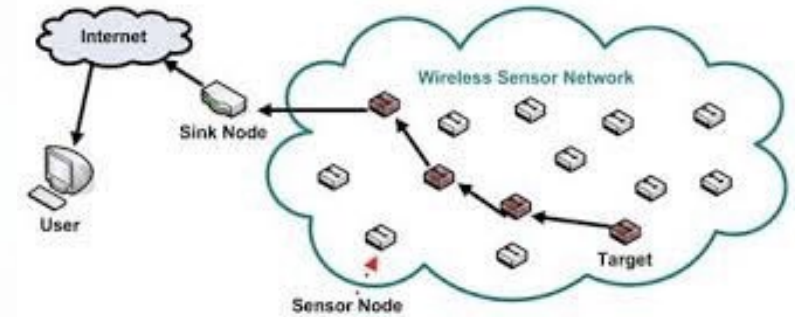


Overview of Security/Privacy Architectures

Application layer	IoT application	Intelligent logistics security	Smart home security	Remote medical security	Smart grid security	Intelligent traffic security	Environmental monitoring security		Other applications security
	Application support layer	Middleware technology security	Service support platform security	Cloud computing platform security	Information development platform security		Other support platform security		
Transportation layer	Local area network	Local area network security							
	Core network	Internet security				3G security		Other networks security	
	Access network	Ad hoc security	GPRS security	WIFI security					
Perception layer	Perception network	RFID Security	Protocol security	WSN security	Routing protocol security	Fusion security		MEMS security, NEMS security, GPS technology security	
			Base station security		Cryptographic algorithms	Sensor +RFID Reader security	RFID +WSN security		
	Perception node		Reader security		Key management	Sensor+Tag security			
			Tag Counterfeit security		Node trust management	Sensor tag security			
			Tag encode security						

Perception Layer

- Information collection layer
 - RFIDs, Wireless sensor networks (WSNs), etc.
- Most vulnerable part
 - Number of devices is large
 - Devices are mostly physically unattended
- What can go wrong?
 - Unauthorized access to raw data
 - Infiltration to network
 - Malicious data injection



Issues of RFIDs and Solutions

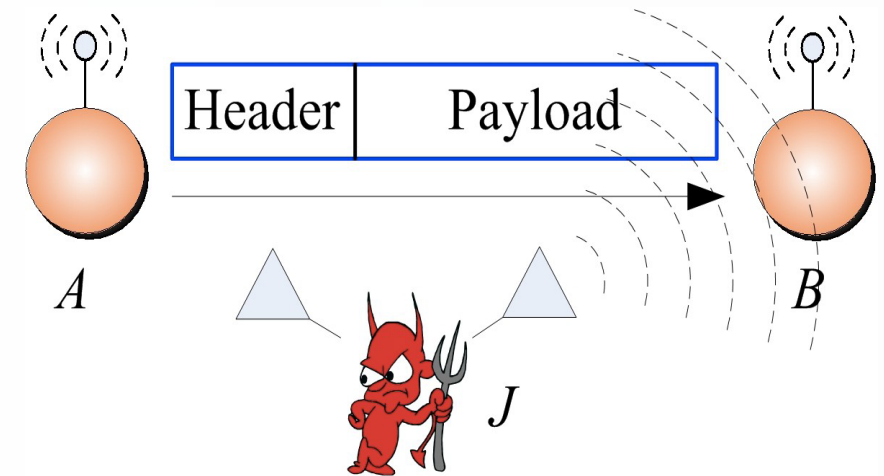
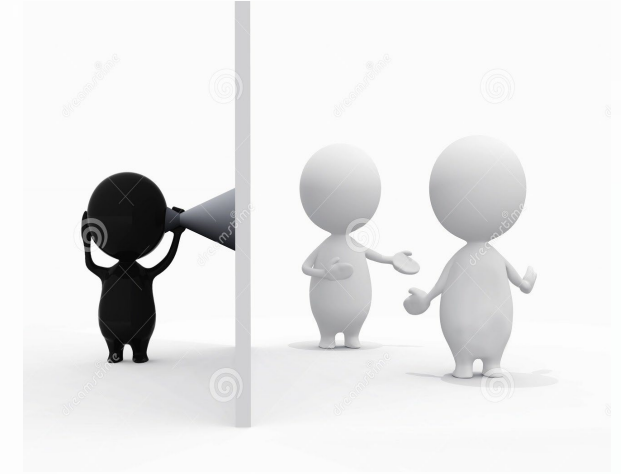
- Potential issues:
 - Non--uniform coding
 - UID (Universal Identification) standards supported by Japan
 - EPC (Electronic Product Code) standard supported by EU
 - Tag reading problems
 - Exposing tag/data
 - Conflict collision
 - Multiple tags in the scope of one reader
 - Multiple readers trying to read a tag
 - Tag vs. reader anti--collusion algorithms

RFID Privacy Issues

- Privacy issues:
 - Lightweight solutions because of limited resources
 - Data privacy
 - Physical--based: deactivation kill commands, block tags, clip tags, etc.
 - Password--based: hash locks, random hash locks, anonymous ID, etc.
 - Location privacy:
 - When a reader equipped with vehicle GNSS information reads a tag's information, it can easily obtain the approximate location information of the tag according to its effective operational range

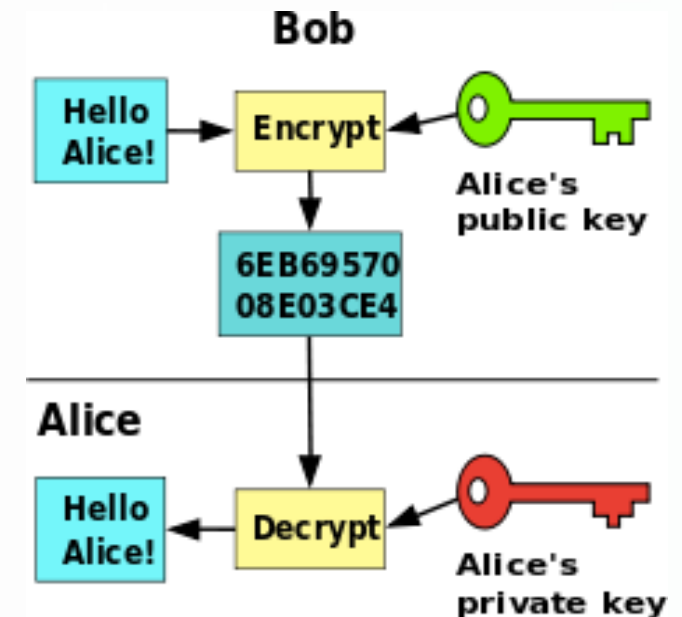
Issues of WSNs

- Issues with WSNs
 - Limited resources (computation power, storage, narrow sensing range)
 - Number of nodes
 - They might collect/transmit sensitive data
- Possible attacks:
 - Eavesdropping
 - Malicious routing
 - Message tampering



Possible Solutions for WSN Security/ Privacy Issues

- **Data encryption**
- Symmetric encryption
 - Asymmetric one are too computationally complex
 - Symmetric one is simpler with less calculation
 - Not scalable (key exchange algorithm is too complex)
 - Key confidentiality problem (nodes are physically unattended)
- Public-key encryption
 - Each node has its own private key and base station's public key
 - Base station has the public keys of all nodes
 - Good scalability
 - Convenient for node authentication
 - Rabin's Scheme, NtruEncrypt and Elliptic Curve Cryptography
 - Two research aspects:
 - Custom, low power HW for the computation work
 - Algorithm design to reduce the amount of computation

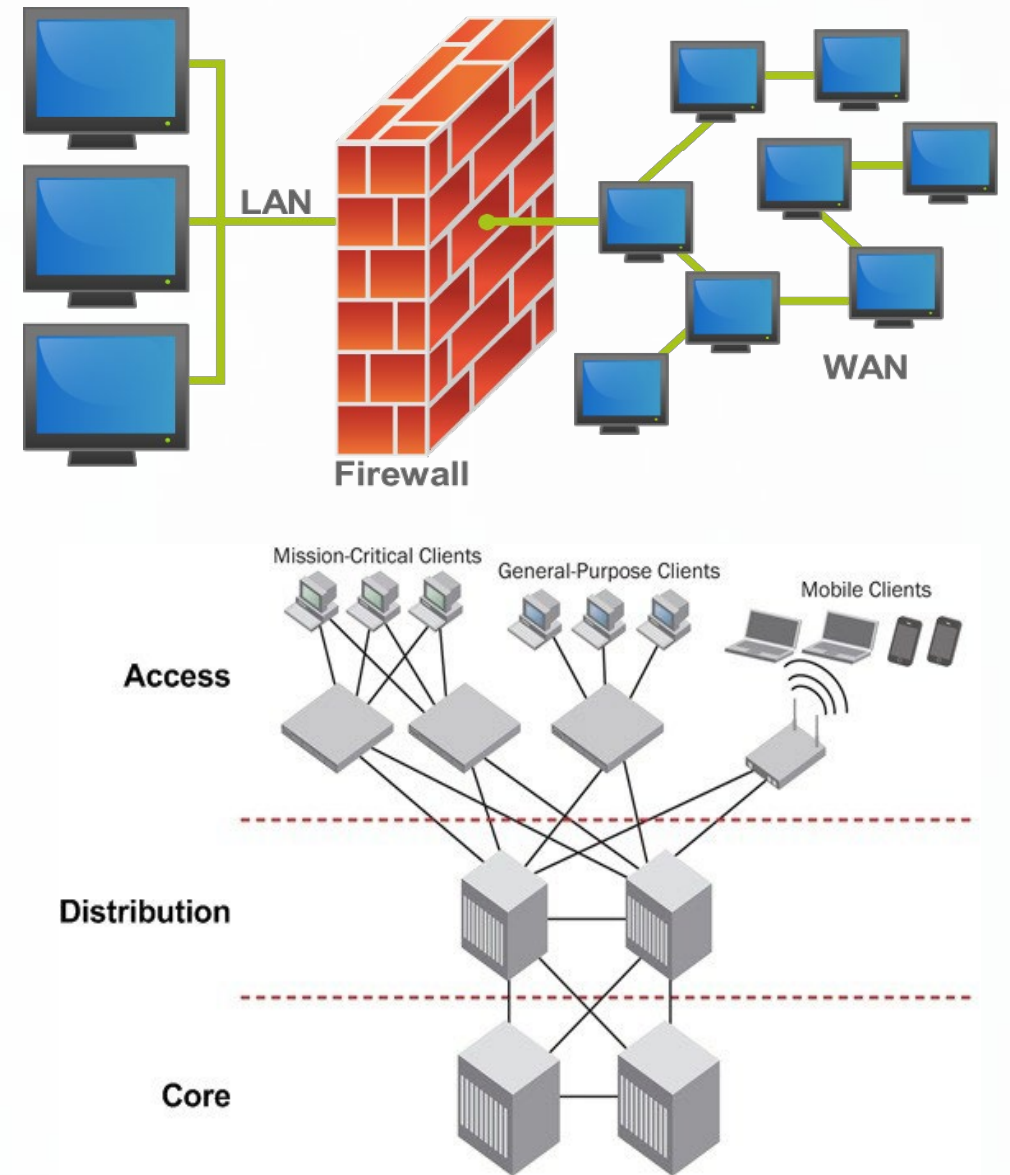


Possible Solutions for WSN Security/ Privacy Issues

- **Key management**
 - Key generation, *distribution*, storage, updating and destruction process
- **Secure routing protocols**
 - Data security can be guaranteed with encryption
 - Remaining problem: authentication of routing information
 - Possible attacks
 - Wormhole attack, Sinkhole attacks, DoS attacks
 - Solution space is not complete
 - Tradeoff between security of the routing vs. energy consumption

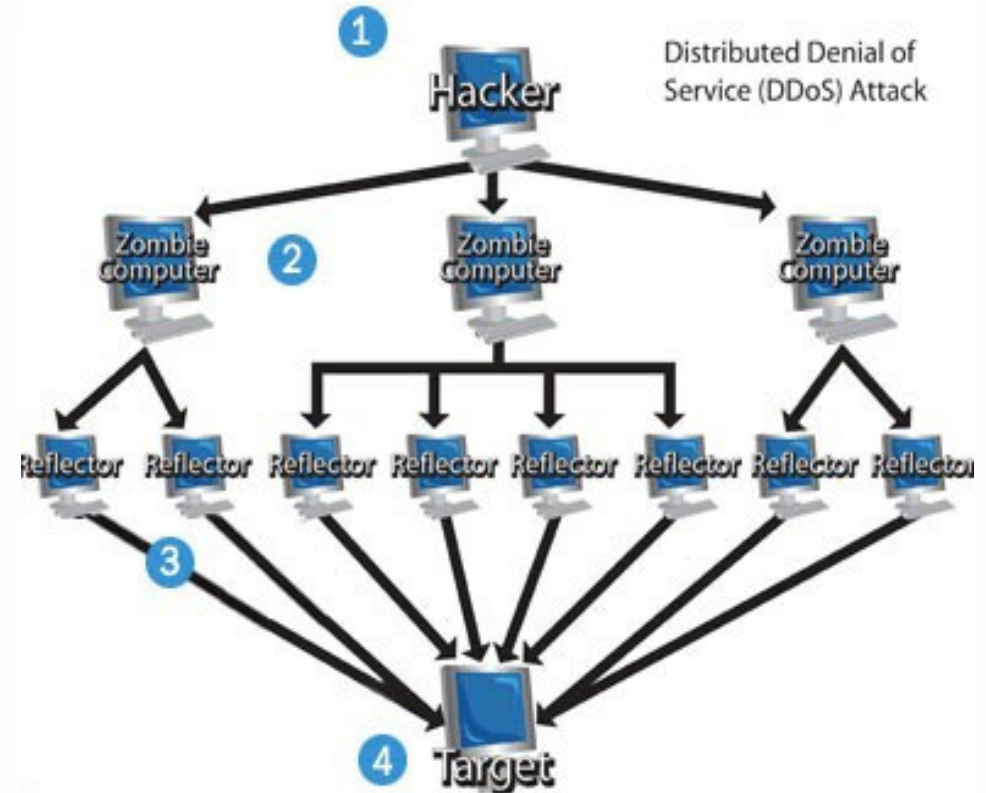
Transportation Layer

- Information transmission between application and perception layer
- Three sub-layers
 - Access network
 - Core network
 - Local area network (LAN)
- What can go wrong?
 - Similar to perception layer
 - Unauthorized access to raw data
 - Infiltration to network
 - Malicious data injection



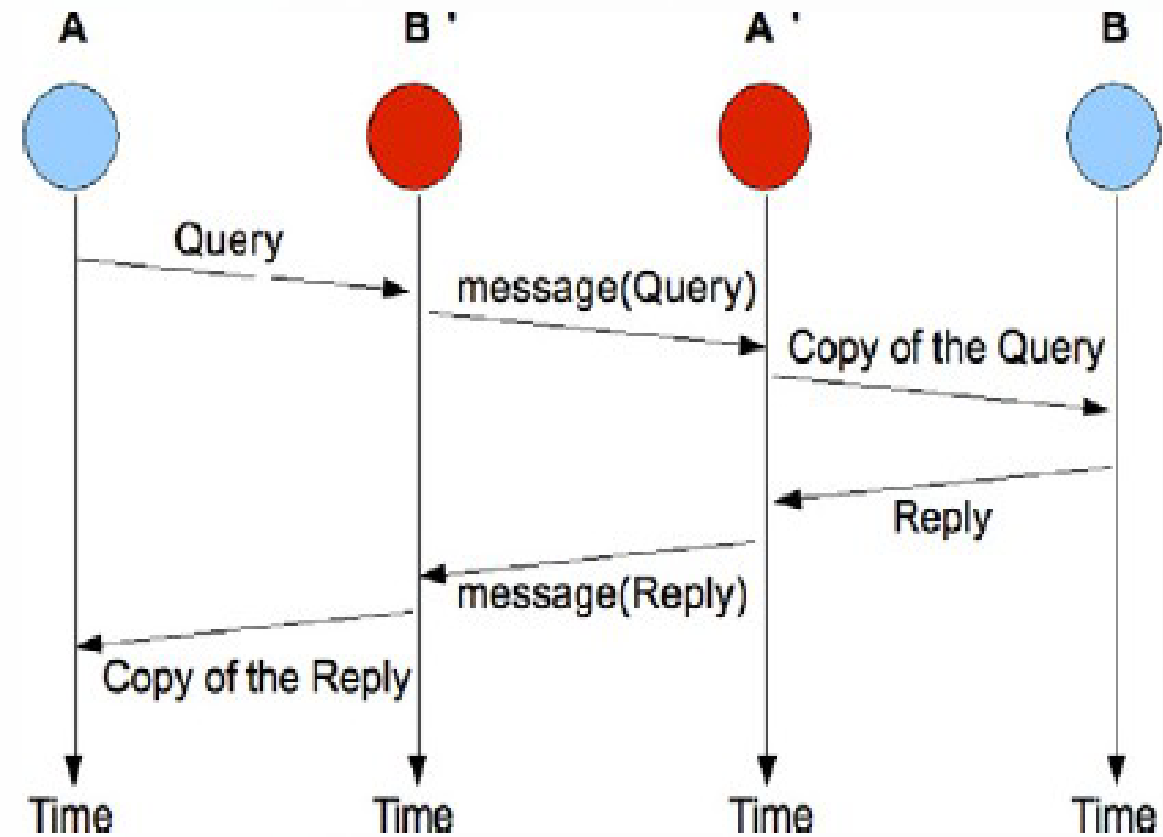
Distributed Denial of Service (DDoS) Attack

- A denial-of-service (DoS) attack is an attempt to make a machine or network resource unavailable to users, to temporarily or indefinitely interrupt or suspend services
- A distributed denial-of-service (DDoS) is where the attack source is more than one, often thousands of, unique IP addresses



Man in the Middle Attack

- The attacker secretly relays and possibly alters the communication between two parties who believe they are directly communicating with each other
 - Active eavesdropping
- The attacker must be able to intercept all relevant messages passing between the two victims and inject new ones
- Solutions:
 - authentication techniques
 - Secure DNS extensions
 - Public key infrastructures
 - Stronger mutual auth.

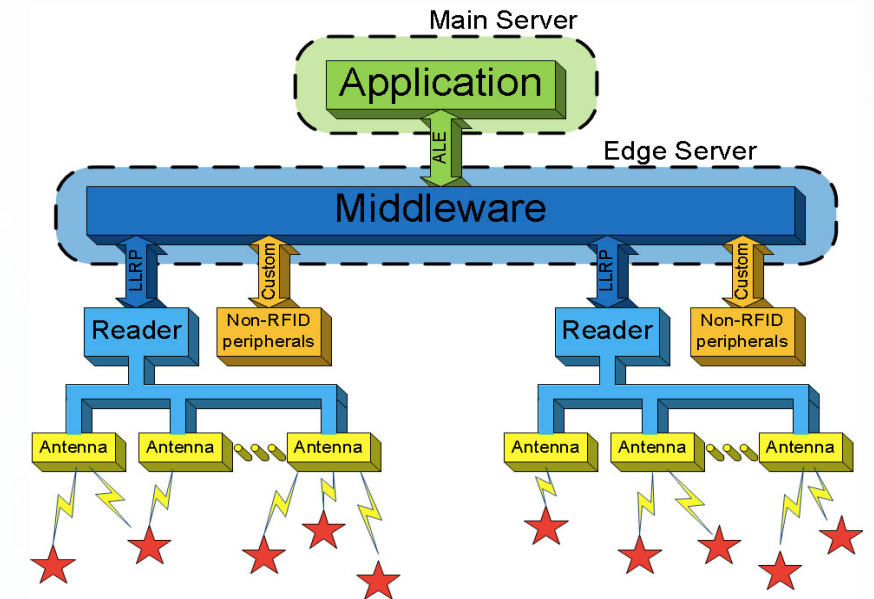


Common Issues and Solutions

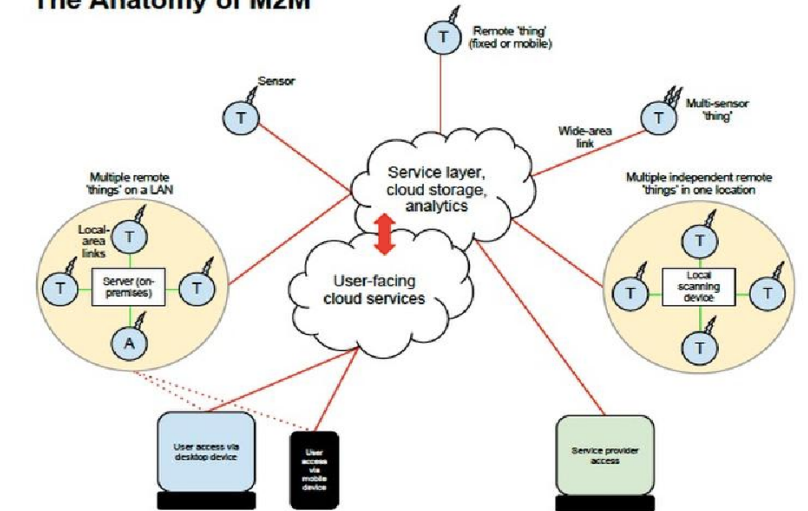
- Wireless network
 - Access attacks, malicious phishing, DDos/Dos attacks
 - Access control: WPA, encryption, authentication technology, etc.
 - Network encryption: TKIP and AES
- Ad hoc network
 - Illegal node access issues (authorization/authentication)
 - Data security issues (authentication)
 - Network routing security issues (encryption)
- 3G/4G network
 - Attacks: user information leakage, data incompleteness, unlawful attacks
 - Possible solutions: key management mechanism, data origin authentication and data encryption

Application Layer

- **Application support layer**
 - Intelligent computation and data processing
 - Middleware, M2M, cloud computing platform and service support platform
- Issues: access control, privacy protect, user authorization, data integrity, real-time availability
- Privacy preserving methods: k-anonymity, data conversion, data randomization, ensuring secure code resources, etc.



The Anatomy of M2M



k-Anonymity

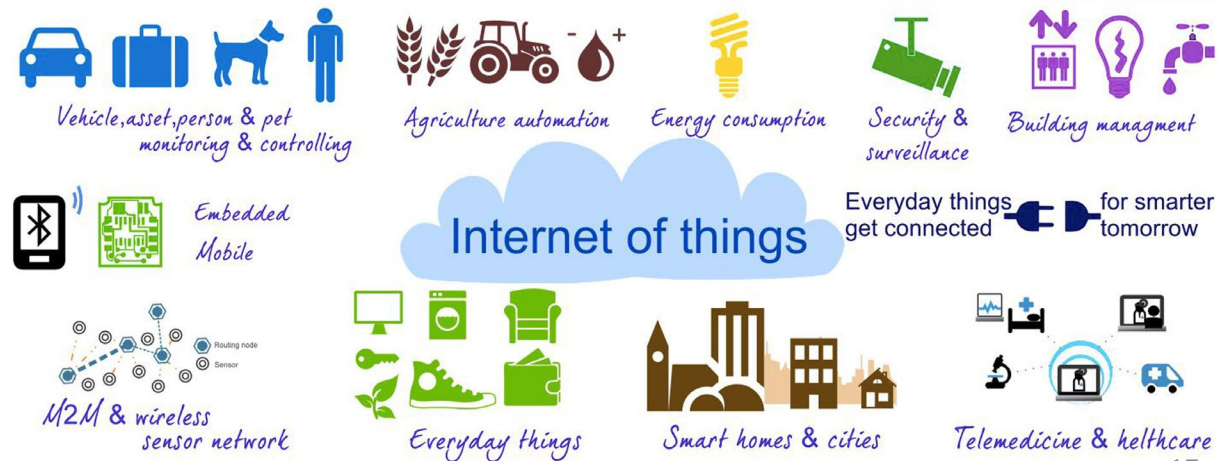
- **Problem:** Given person--specific field--structured data, produce a release of the data with scientific guarantees that the individuals who are the subjects of the data cannot be re--identified while the data remain practically useful
- A release of data is said to have the k--anonymity property if the information for each person contained in the release cannot be distinguished from at least k--1 individuals whose information also appear in the release.

Name	Age	Gender	State of domicile	Religion	Disease
Ramsha	29	Female	Tamil Nadu	Hindu	Cancer
Yadu	24	Female	Kerala	Hindu	Viral infection
Salima	28	Female	Tamil Nadu	Muslim	TB
sunny	27	Male	Karnataka	Parsi	No illness
Joan	24	Female	Kerala	Christian	Heart-related
Bahuksana	23	Male	Karnataka	Buddhist	TB
Rambha	19	Male	Kerala	Hindu	Cancer
Kishor	29	Male	Karnataka	Hindu	Heart-related
Johnson	17	Male	Kerala	Christian	Heart-related
John	19	Male	Kerala	Christian	Viral infection

Name	Age	Gender	State of domicile	Religion	Disease
*	20 < Age ≤ 30	Female	Tamil Nadu	*	Cancer
*	20 < Age ≤ 30	Female	Kerala	*	Viral infection
*	20 < Age ≤ 30	Female	Tamil Nadu	*	TB
*	20 < Age ≤ 30	Male	Karnataka	*	No illness
*	20 < Age ≤ 30	Female	Kerala	*	Heart-related
*	20 < Age ≤ 30	Male	Karnataka	*	TB
*	Age ≤ 20	Male	Kerala	*	Cancer
*	20 < Age ≤ 30	Male	Karnataka	*	Heart-related
*	Age ≤ 20	Male	Kerala	*	Heart-related
*	Age ≤ 20	Male	Kerala	*	Viral infection

Applications

- Integrated or individual specific application business
- Need specific solutions that cannot be handled by other layers
- Privacy becomes a big issue
 - Data becomes meaningful
 - Location privacy
 - Query privacy
 - Location camouflage, anonymous space, space encryption, etc.



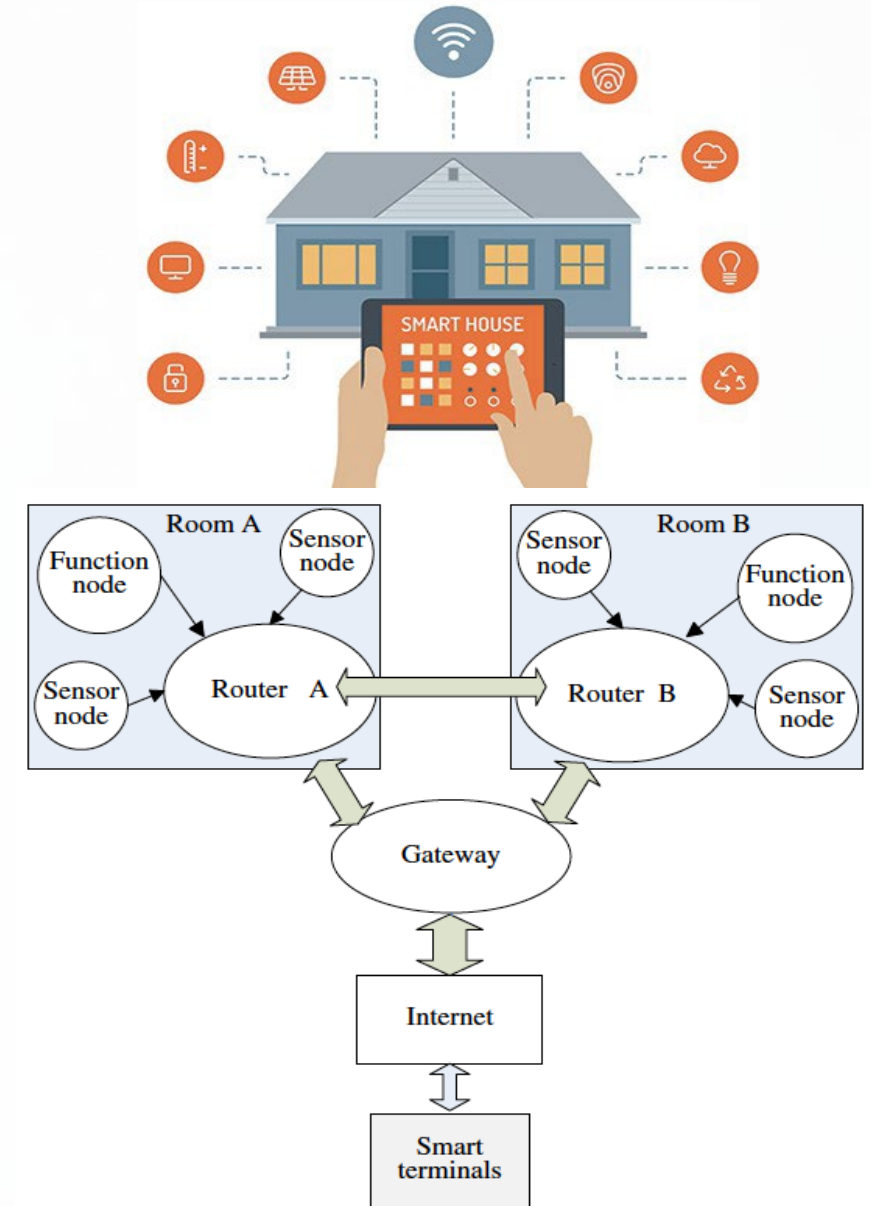
App 1: Intelligent Transportation

- Logistics industry
- Vehicle tracking, inventory management, process monitoring
- Biggest enabler: RFID tags
 - Prone to hacking and security attacks
 - **Problem: information leakage**
 - Solutions: Data encryption, not storing/sending sensitive data through RFIDs
- Monitoring the items
 - Zigbee has been used as a low-rate, low power, low cost, self-configuration and flexible network topology solution

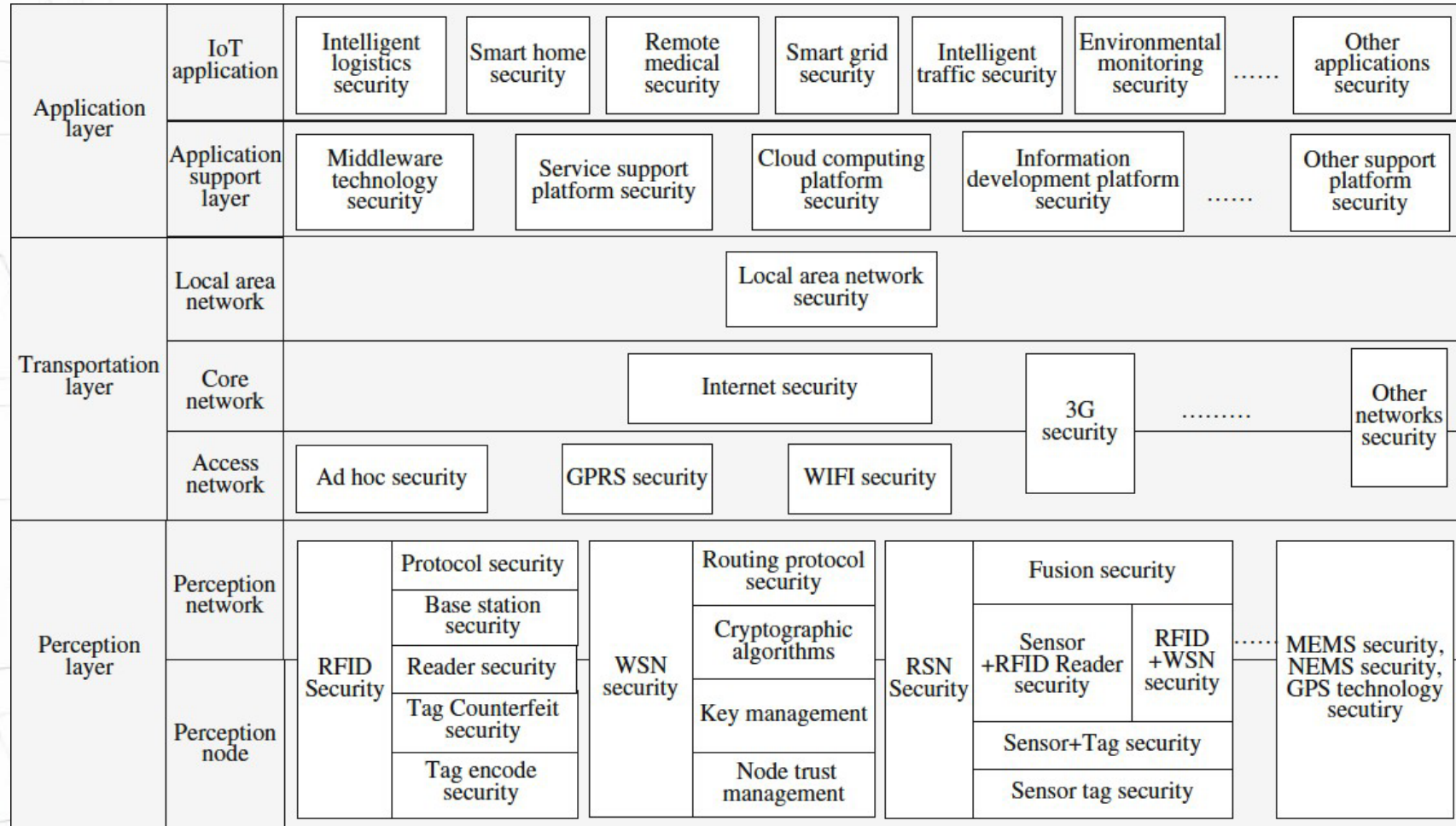


App2: Smart Home

- Sensors, Internet and intelligent control
- The mainly applied technology: network control, communications and mobile terminal technology
- Possible problems:
 - Hacking control units
 - Information leakage vs. sharing unintended information (privacy)
 - Recovering occupancy patterns or habits
- Solutions depend on the HW and the SW layer that they are applied on



Overview of Security/Privacy Architectures



Jing et al. Security of the Internet of Things: perspectives and challenges.

Summary and Open Research Issues

- IoT has different security/privacy needs than traditional networks
 - Computation power, less secure communication channels, no high level OS support, etc.
- The security/privacy issues/solutions depend on the hierarchical structure of the IoT
 - Perception – transportation – application layer
 - No one solution that can target every layer!
- Open research issues:
 - Overall security/privacy architecture for the entire IoT system
 - Computationally lightweight solutions
 - Efficient solutions for massive heterogeneous data

References

- <https://qiro.com/ecs6264/lectures/>
- <https://cseweb.ucsd.edu/classes/wi16/cse291-c/>



ULAB
UNIVERSITY OF LIBERAL ARTS
BANGLADESH

Thank You

atanu.Shuvam@ulab.edu.bd

University of Liberal Arts Bangladesh
Department of Computer Science & Engineering